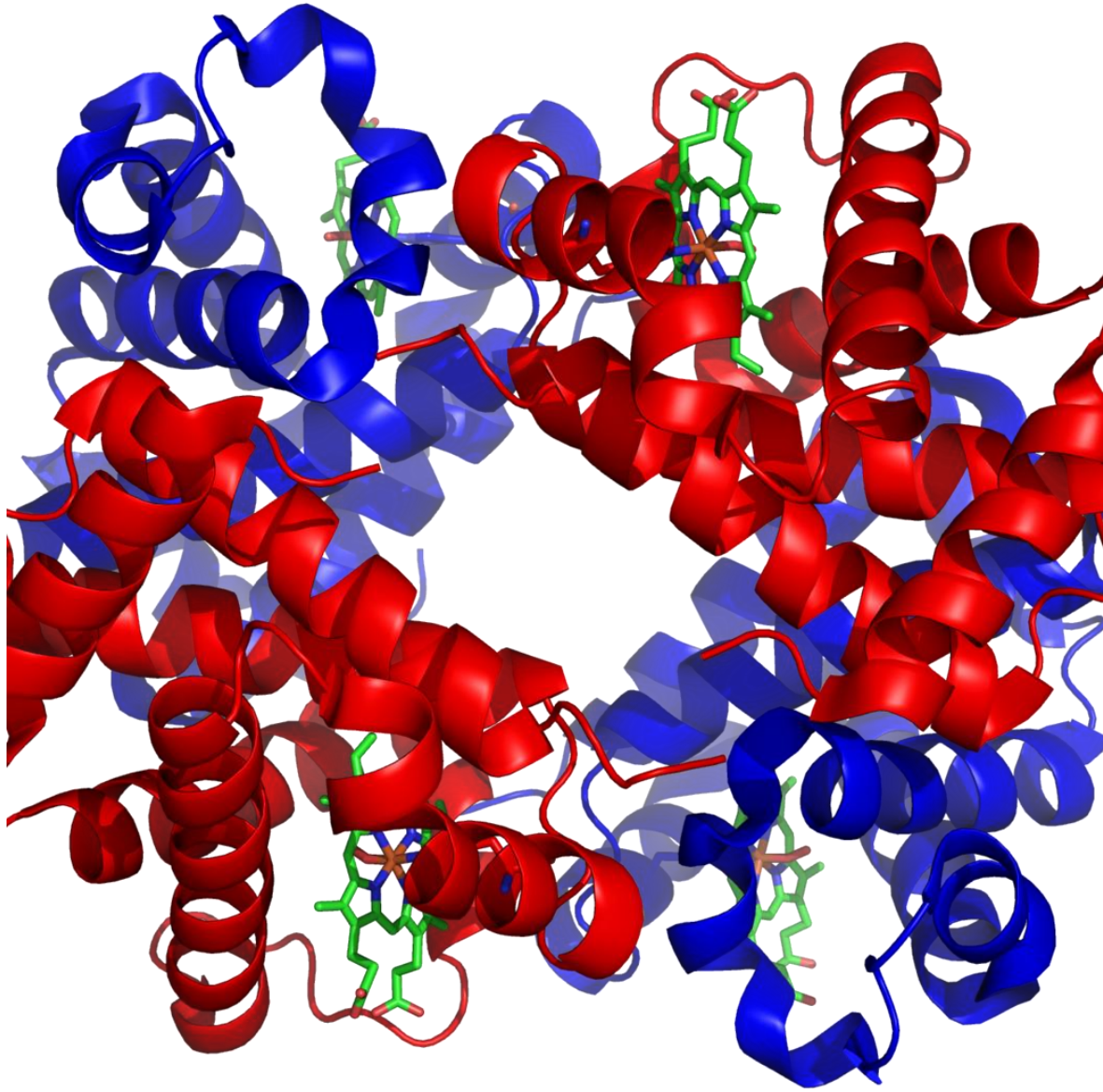
DP Biology – B1.2

Proteins

- How does the order of amino acids produce such a variety of protein forms and functions?
- How do chemical and physical conditions affect protein molecules?

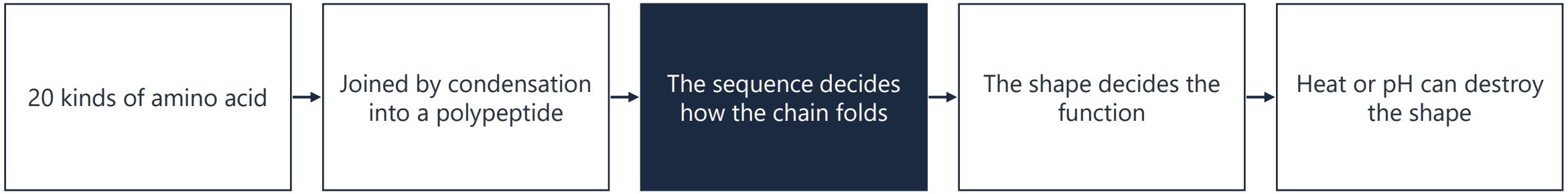


SL statements B1.2.1–5. Textbook pp. 205–210.
Image: model of haemoglobin. Zephyris, CC BY-SA 3.0.

Objectives for each class

Class	Statements	By the end of the class, you can...
1	B1.2.1–2	Draw a generalized amino acid, and show how condensation forms a dipeptide and then a polypeptide
2	B1.2.3–4	Explain dietary needs for amino acids, and why the variety of peptide chains is almost limitless
3	B1.2.5	Explain how high temperature and changes in pH denature proteins

The big picture: from sequence to function



Proteins do most of the work in cells: they speed up reactions, carry oxygen, fight infection, give strength to skin and move muscles. All are built from the same 20 kinds of **amino acid**.

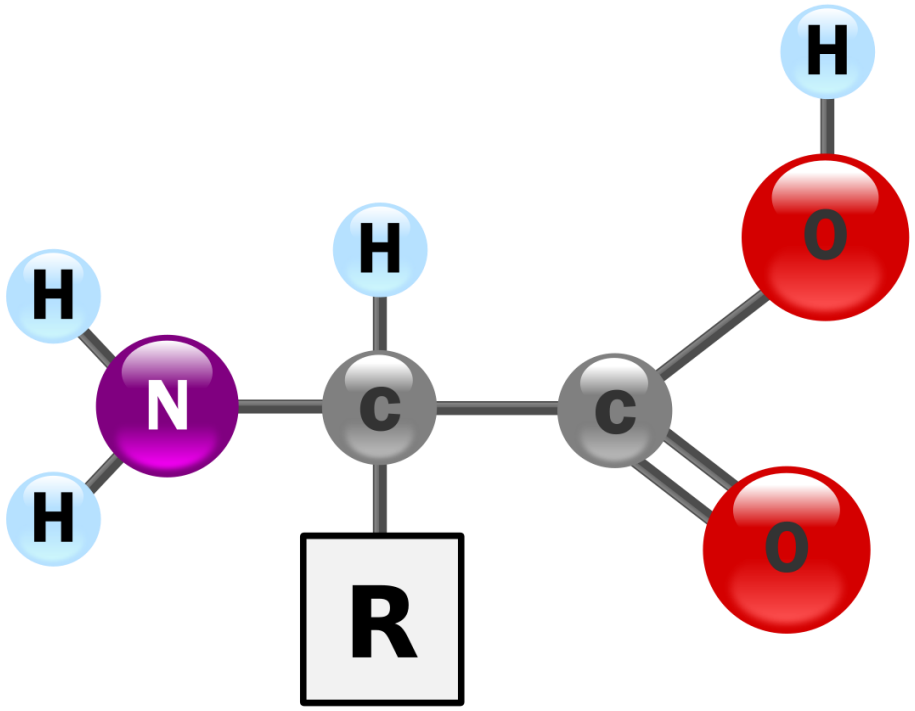
What makes each protein different is the **order** of its amino acids. That order makes the chain fold into one particular shape, and the shape lets it do one particular job.

Big question for this chapter

How can the same 20 building blocks make an enzyme, an antibody, a hormone and a hair?

The structure of an amino acid

B1.2.1



General structure of an amino acid. YassineMrabet, public domain.

Every amino acid has a central **alpha carbon** with four groups attached:

an **amine group** (-NH_2),

a **carboxyl group** (-COOH),

a **hydrogen atom** (-H),

and an **R-group**, which is different in each of the 20 amino acids.

Exam tip

Be ready to draw a generalized amino acid with R for the side chain. You do not need to learn the R-groups of particular amino acids.

R-groups make each amino acid different

B1.2.1

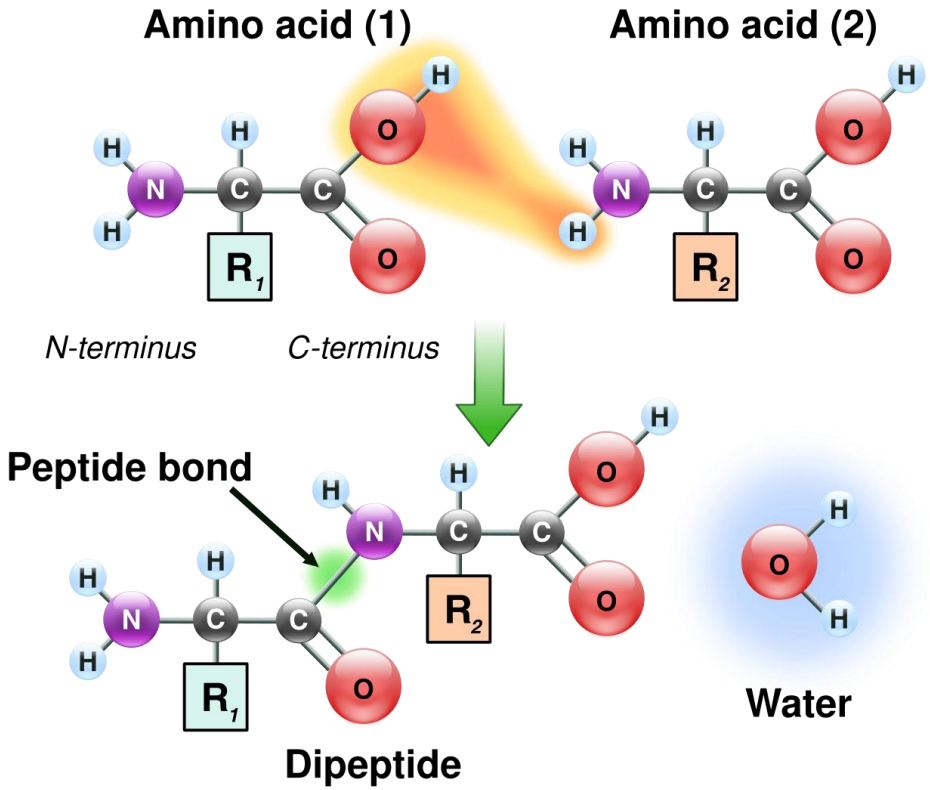
Type of R-group	Examples	What it does in a protein
Non-polar	Alanine (a methyl group), leucine	Hydrophobic: tends to be buried inside the folded protein
Polar	Serine (-CH_2OH)	Hydrophilic: forms hydrogen bonds with water and other groups
Charged (acidic or basic)	Aspartic acid, lysine	Forms ionic attractions; affected by pH
Smallest	Glycine (-H)	Lets the chain bend tightly

Key idea

Because R-groups differ in size, charge and polarity, each sequence folds in its own way.

Condensation joins two amino acids

B1.2.2



Two amino acids form a dipeptide. YassineMrabet, public domain.

The **carboxyl group** of one amino acid reacts with the **amine group** of the next. An -OH and an -H are removed as **water**.

The new covalent bond between C and N is a **peptide bond**. Two amino acids joined in this way make a **dipeptide**.

Word equation

amino acid + amino acid → dipeptide + water

How to draw a dipeptide

B1.2.2

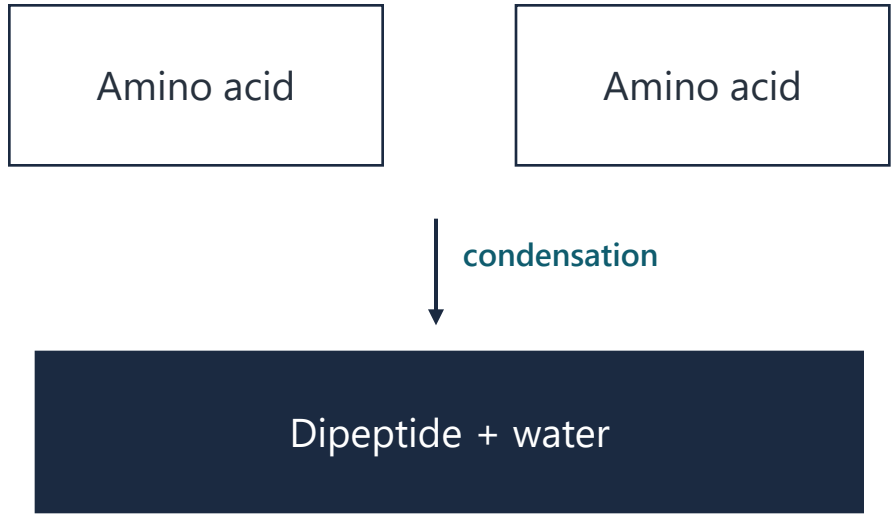
1. Draw two amino acids side by side, both with the amine group on the left and the carboxyl group on the right.

2. Circle the -OH of the first carboxyl group and an -H of the second amine group.

3. Redraw them joined: the C of the first now bonds to the N of the second. Label this the **peptide bond**.

4. Write **+ H_2O** on the products side.

5. Label the amine end and the carboxyl end of the dipeptide.



Try it

Use molecular model kits to build two amino acids and join them, then draw what you built.

From dipeptides to polypeptides

B1.2.2

Each new amino acid is added by another condensation reaction, making a chain called a **polypeptide**. In cells this happens on **ribosomes**, following the code in mRNA, and each step releases one water molecule.

Polypeptide	Number of amino acids	Function
Insulin (two chains)	21 and 30	Hormone that lowers blood glucose
Haemoglobin (four chains)	141 and 146 each	Carries oxygen in red blood cells
Titin	About 34 000	Elastic spring in muscle

Common mistake

A polypeptide and a protein are not the same thing. A protein is one or more polypeptides folded into a working shape.

Check-in: amino acids and peptide bonds

B1.2.1–2

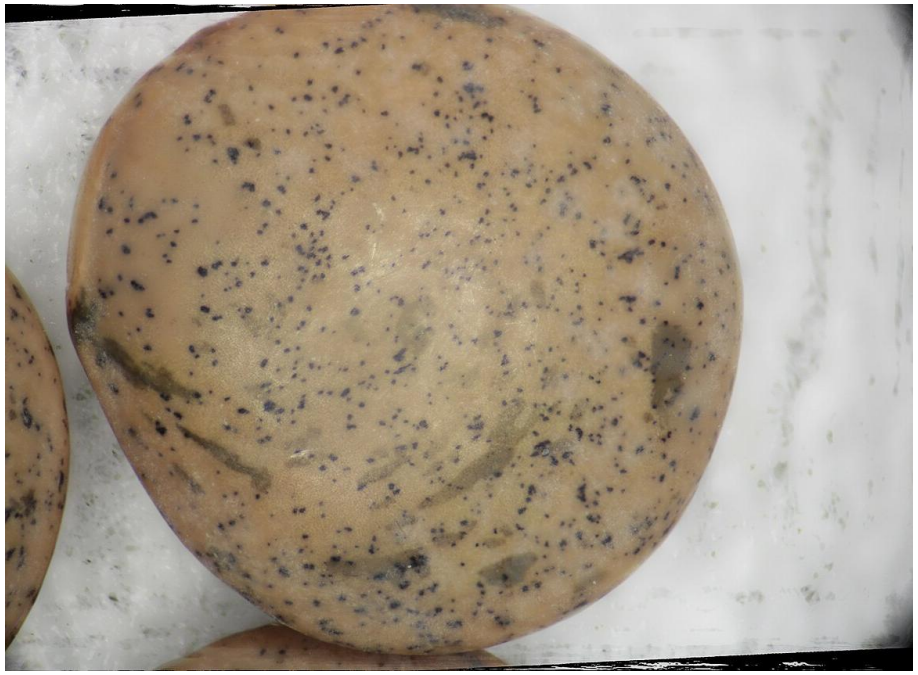
1. Name the four groups attached to the alpha carbon of an amino acid.
2. Draw a generalized amino acid.
3. State the name of the bond that forms between two amino acids.
4. Outline how a dipeptide forms.
5. Calculate the number of water molecules released when 50 amino acids join in one chain.

How to use this

Answer without your notes, then compare with a partner.

Essential and non-essential amino acids

B1.2.3



Lentils: a plant source of protein. Bean appreciator, CC0.

Humans need all 20 amino acids to make their proteins.

Non-essential amino acids can be made in the body from other amino acids.

Essential amino acids cannot be made, so they must come from food. For adults there are nine.

If one essential amino acid runs short, protein synthesis slows, even if the others are plentiful.

Discussion

Why might a shortage of just one essential amino acid limit how much protein a person can make?

Plant-based diets and complementary proteins

Food group	Usually low in	Good partner
Cereals (rice, wheat, maize)	Lysine	Pulses such as beans or lentils
Pulses (beans, lentils, peas)	Methionine	Cereals, nuts or seeds
Soya and quinoa	Contain all the essential amino acids	

Meat, eggs and dairy usually contain all the essential amino acids. Vegans can get them all by eating a **variety of plant foods**: rice with beans, or bread with hummus, supply what each lacks alone.

Link

Diet and health; C4.2: energy and nutrient flow through food chains.

An almost limitless variety of polypeptides

B1.2.4

Chain length	Calculation	Sequences
1	20	20
2	20 × 20	400
3	20^3	8000
4	20 to the power 4	160 000
6	20 to the power 6	64 million

There are 20 amino acids in the genetic code, and any amino acid can follow any other.

For a chain of n amino acids there are **20 to the power n** possible sequences, and chains range from a few to thousands of amino acids long.

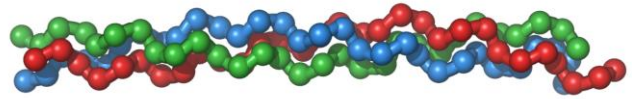
So the possible variety is, in practice, limitless.

Key idea

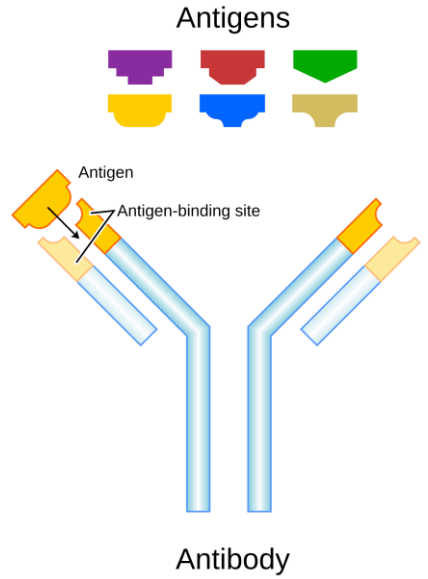
A chain of 400 amino acids has 20 to the power 400 possible sequences. Only a tiny fraction of these has ever been made by any organism.

Examples of proteins and their functions

Protein	Function
Haemoglobin	Carries oxygen in red blood cells
Antibodies	Bind to specific antigens on pathogens
Rhodopsin	Light-absorbing pigment in the retina
Collagen	Rope-like strength in skin, tendons and bone
Keratin	Tough structure of hair, nails and feathers
Insulin	Hormone that signals cells to take in glucose



Collagen. CC BY-SA 3.0.

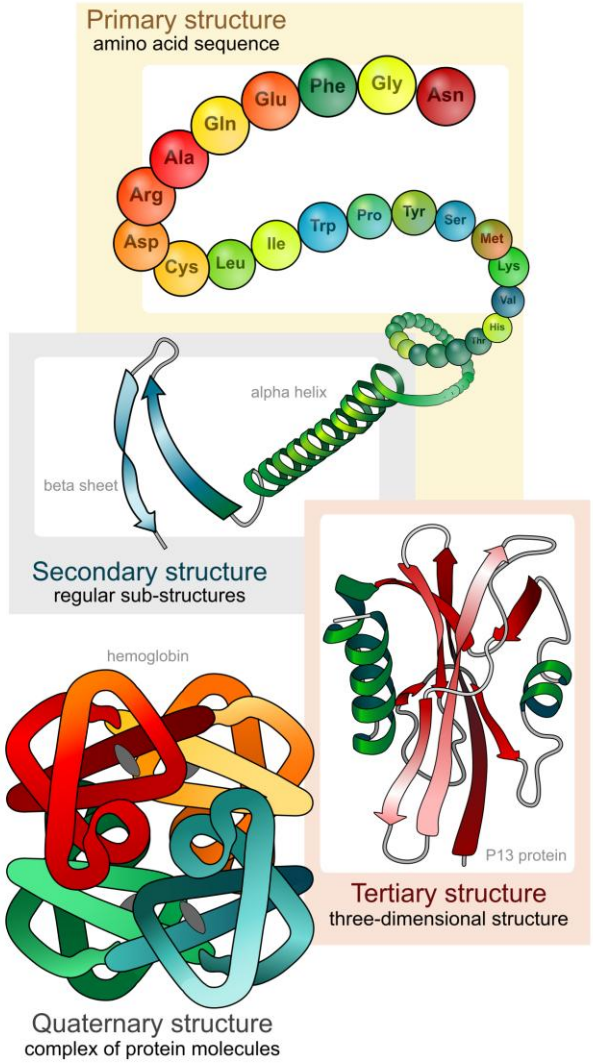


An antibody. Fvasconcellos, public domain.

Big question: same 20 building blocks, different sequences, different shapes, different jobs.

Sequence determines shape

B1.2.4



Levels of protein structure. LadyofHats, public domain.

As a polypeptide is made, its R-groups interact with each other and with water. Non-polar R-groups cluster in the middle, away from water; charged and polar ones attract each other.

So a particular sequence **always folds the same way**, giving a particular three-dimensional shape, and that shape lets the protein do its job.

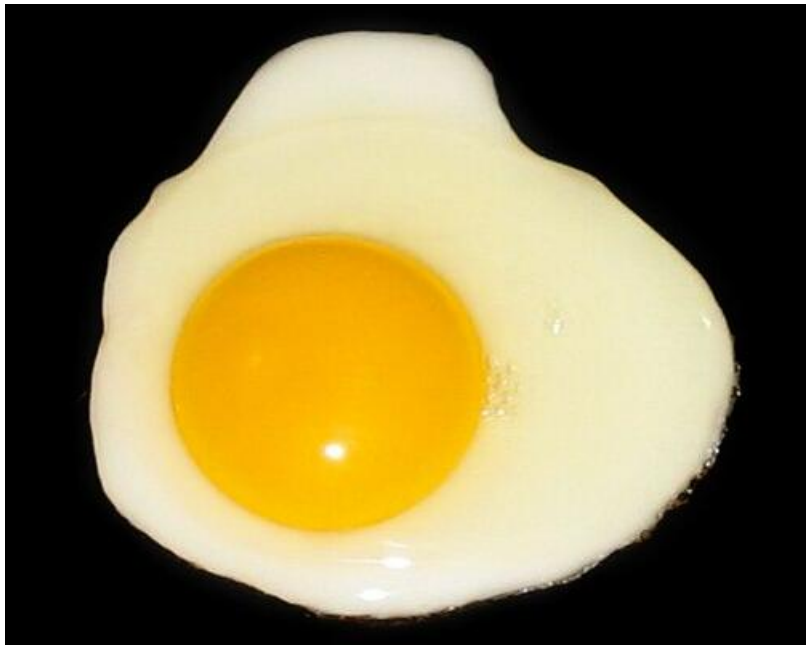
Change one amino acid and the shape, and the function, may change too.

Link

HL B1.2.7–11: levels of structure and the bonds that hold them. D1.2.11: how a mutation can change a protein (sickle cell).

Denaturation by heat

B1.2.5



Cooked egg: heat has denatured the albumin. David Benbennick, public domain.

Heat makes the atoms in a protein vibrate more. Above a certain temperature, the weak bonds and interactions that hold its shape break, and the chain **unfolds**.

The protein has lost its three-dimensional shape: it is **denatured**. It can no longer do its job, and usually cannot refold.

Egg white turns from clear and runny to white and solid when it is cooked because its proteins denature and tangle together.

Common mistake

Denaturation does not break peptide bonds, so the amino acid sequence is unchanged. Only the folded shape is lost.

Denaturation by pH

B1.2.5

Some R-groups carry a charge and are held in place by attraction to groups of opposite charge. Adding or removing H^+ ions (changing the pH) changes those charges, so the attractions break and the protein changes shape.

Protein	Works best at pH	What this shows
Pepsin (stomach protease)	About 2	Adapted to the strong acid in the stomach
Salivary amylase	About 7	Stops working once it reaches the stomach
Trypsin (small intestine)	About 8	Adapted to the alkaline conditions there

Key idea

Each protein keeps its shape only within a range of pH and temperature that suits where it works.

Working with data: heating egg white

B1.2.5

Egg white solution was held at each temperature for 10 minutes, then its cloudiness was measured with a colorimeter (higher absorbance = more denatured protein). These data are invented for practice; the pattern is realistic.

Temperature / °C	Absorbance
40	0.02
50	0.03
60	0.21
70	0.64
80	0.95
90	0.97

1. Describe the effect of temperature on absorbance.
2. Explain the change between 50 and 80 °C.
3. Suggest why absorbance hardly changes above 80 °C.
4. State two variables that should be controlled.

Check-in: diet, variety and denaturation

B1.2.3–5

1. Distinguish between essential and non-essential amino acids.
2. Explain why vegans are advised to eat a variety of plant protein foods.
3. Calculate the number of possible sequences for a peptide of 3 amino acids.
4. Explain why a boiled egg cannot become raw again.
5. Explain why salivary amylase stops working in the stomach.

How to use this

Write full sentences, then mark with a partner.

Common mistakes to avoid

Not this	But this
A peptide bond joins two R-groups.	A peptide bond joins the carboxyl carbon of one amino acid to the amine nitrogen of the next.
Hydrolysis forms a dipeptide.	Condensation forms it and releases water; hydrolysis breaks it.
Polypeptide and protein mean the same.	A protein is one or more polypeptides folded into a working shape.
Denaturation breaks the protein into amino acids.	Denaturation changes the shape; peptide bonds and sequence stay the same.
Essential amino acids are the most important ones.	Essential means they must come from the diet; the body needs all 20.

Check that you can...

1. **Draw** a generalized amino acid and **label** its groups
2. **Draw** a dipeptide and **write** the word equation for its formation
3. **Distinguish** between essential and non-essential amino acids, and **discuss** plant-based diets
4. **Calculate** the number of possible peptide sequences, and **explain** why variety is limitless
5. **Outline** examples of proteins and their functions
6. **Explain** how sequence determines shape
7. **Explain** how heat and pH denature proteins

Exit activity: 5 minutes

On your own and without notes. Hand this in as you leave.

1. Name the four groups attached to the alpha carbon of an amino acid.
2. Calculate the number of possible sequences for a peptide of 3 amino acids.
3. Explain why a boiled egg cannot go back to being raw.

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